

Little receipts. A clearer picture.

Tran Lam · 08 October 2026 · Flutter 3.47.5 / Dart 3.13.4 · Android + web companion

Students and club treasurers can turn printed receipts into reviewed expense records without a cloud OCR service. PocketReceipt keeps capture, recognition, editing, storage and spending insights in one local-first workflow.

Feature checklist

OK Camera and crop

Live preview, flash toggle, tap focus, framing overlay and adjustable crop edges.

OK Offline recognition

Bundled Google ML Kit Latin model; image import and measured OCR-call latency.

OK Heuristic parsing

Merchant, valid calendar dates and labelled totals; grouped VND amounts and review warnings.

OK Review and lifecycle

Editable amount, merchant, date and category; add, search, filter, edit and delete.

OK Private persistence

SQLite records, receipt JPEGs and 160-pixel thumbnails in app-private storage.

OK Canvas insights

Animated donut and seven-day bars, with category selection and bar inspection.

OK Submission package

Public repository, release APK, real OCR demonstration video and this four-page report.

Scope note: the web companion demonstrates text parsing, local browser persistence and charts. Camera image OCR and SQLite are Android features. The official institutional report template was not supplied; this report follows the required checklist, architecture and screenshots structure.

From pixels to a useful record.

A small, modular Flutter codebase with no server and no third-party chart library.

CAPTURE / CROP

Camera or gallery → EXIF orientation → crop → JPEG

ON-DEVICE OCR

ML Kit Latin recognizer → raw text + elapsed milliseconds

PARSE / REVIEW

Regex candidates → ranked totals → editable confirmation

PERSIST / VISUALIZE

SQLite + private images → calendar summaries → CustomPainter

Module boundaries

domain/ holds the Expense model, categories and deterministic parser. services/ owns ML Kit creation, asynchronous recognition and disposal. data/ conditionally selects Android SQLite/files or web preferences. ui/ contains capture/crop, review, overview/history and animated painters.

Data contract and lifecycle

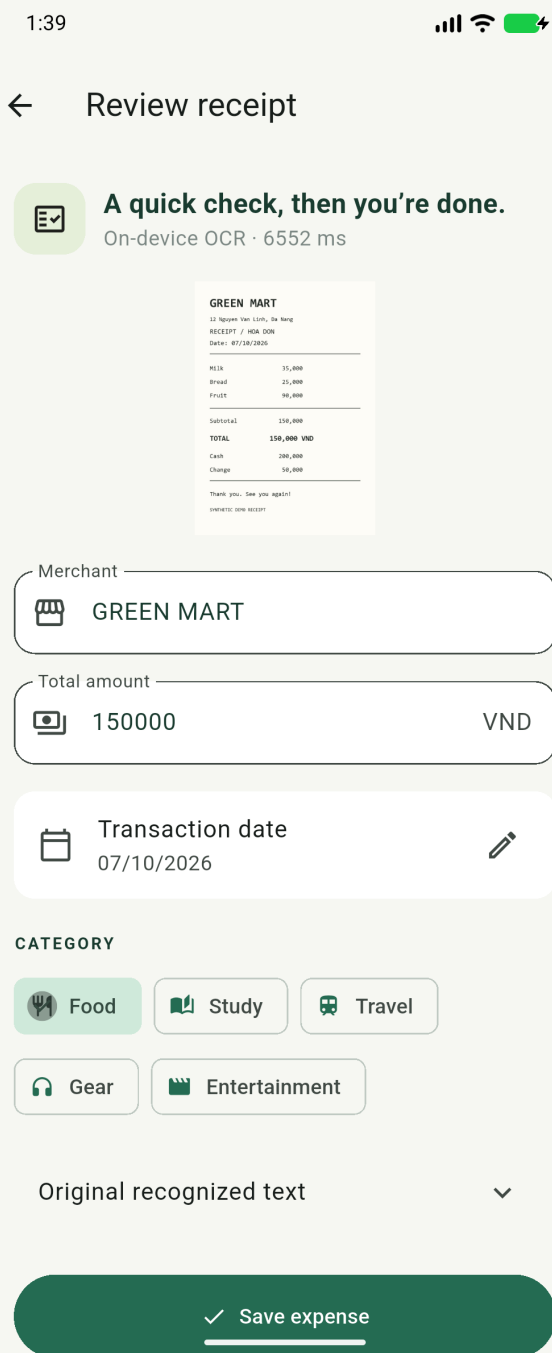
Schema v1: id (primary key), merchant, amount (positive integer VND), date (local ISO date), category, receiptPath and rawText. Saving replaces the same ID during edits. Cancelled scans and deleted expenses remove their cached receipt files. Android backup is disabled.

Parsing decisions

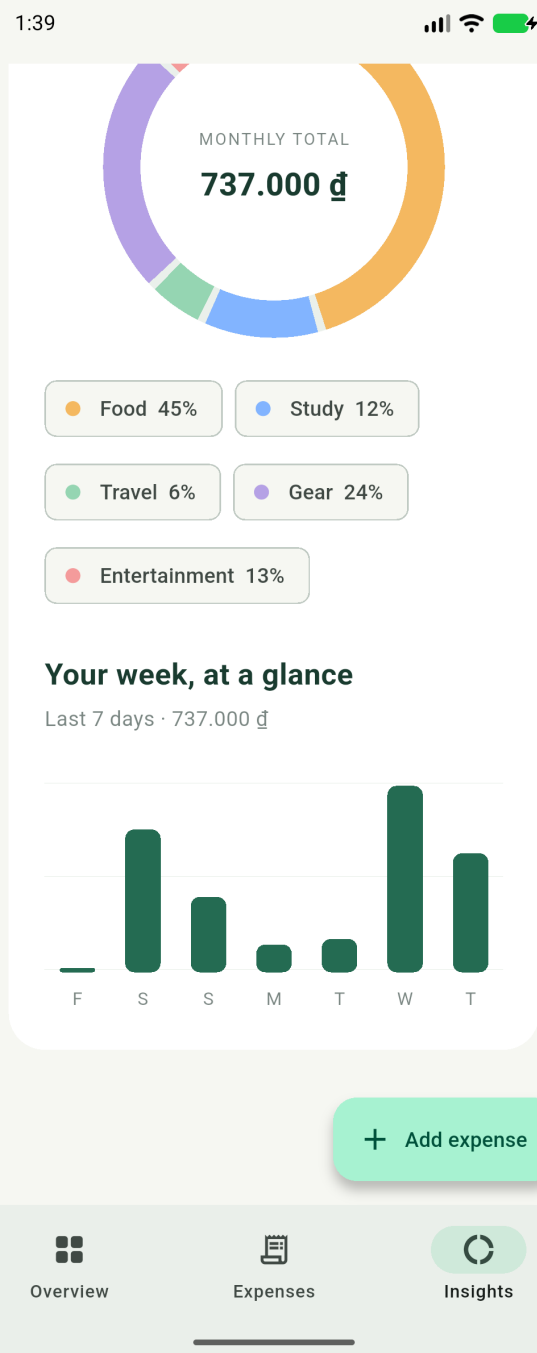
Accents are normalized for matching. Grand-total labels outrank generic totals; subtotal, VAT, cash, discounts and change are excluded. A total on the following line is supported. Invalid dates and unlabelled totals remain unset, and competing totals produce a warning. Keyword categories are suggestions; every extracted field remains editable.

Evidence from the Android app.

Actual emulator screenshots. The receipt fixture is synthetic; recognition uses the native ML Kit model.



01 Actual OCR result and review



02 Interactive canvas insights

Tested, measured, and reviewable.

Verification distinguishes implemented behavior from performance targets and physical-device coverage.

Automated verification

29 tests passed. Coverage includes VND separators/currency suffixes, invalid amounts/dates, leap years, Vietnamese and split-line totals, spatial OCR row reconstruction, missing/ambiguous totals, category suggestions, SQLite insert/update/reopen/delete, review validation and empty charts. flutter analyze completed with no issues. Release APK and production web builds completed.

End-to-end evidence

The final release APK was installed on the Android emulator. Importing the synthetic receipt into native ML Kit returned GREEN MART, 150000 VND and 07/10/2026 correctly, with a displayed OCR-call duration of 6552 ms. Saving and force-stopping/relaunching preserved the expense in SQLite. The 150-second video records this flow and then adds labelled sample data to exercise interactive charts and history.

Performance and remaining limits

The sub-100 ms requirement is treated as a target, not a guarantee. The displayed measurement covers the asynchronous ML Kit call, including platform/model initialization, and excludes camera/crop time. Emulator measurements do not establish physical-phone latency. Camera focus/flash hardware behavior, faint thermal print, skew and varied merchants still require a physical-device test matrix. This version supports integer VND and Latin-script text; category suggestions can be wrong.

Reproduce the build

flutter pub get → flutter analyze → flutter test → flutter build apk --release. Install the APK on Android API 24+. The educational release uses development signing for sideloading; Play Store publication requires a private signing key. See README for web build commands, architecture and privacy.

Public source repository

<https://github.com/it2kvku/pocketreceipt-ocr-expense-tracker>

APK + video + report

<https://github.com/it2kvku/pocketreceipt-ocr-expense-tracker/releases/tag/v1.0.0>

Live web companion

<https://pocketreceipt-demo.onrender.com>

Primary references: developers.google.com/ml-kit/vision/text-recognition/v2/android; pub.dev/packages/google_mlkit_text_recognition; pub.dev/packages/sqlite; api.flutter.dev/flutter/rendering/CustomPainter-class.html.